Table A2. Random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), crashes involving a motorcycle and another vehicle type, Thailand, 2016–2023 (yearly periods), partially constrained temporal structure.

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Constant [serious injury / fatal]				
Constant [2016–2017]	0.399	2.49		
Constant [2018–2019]	0.021	0.14		
Constant [2020–2022]	–0.310	–2.15		
Constant [2023]	–0.023	–0.14		
Variables producing the same parameter value across all periods [2016–2023]				
Vehicle involvement				
One 10-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	1.416	11.73	–0.1942	0.1942
One truck with more than 10 wheels involved (1 if yes, 0 otherwise) [2016–2023]	2.736	7.93	–0.2611	0.2611
(Standard deviation of parameter distribution; 79.8% above zero)	3.279	6.05		
Crash type				
Sideswipe (parallel movement) (1 if yes, 0 otherwise) [2016–2023]	–0.683	–11.34	0.0925	–0.0925
Head-on collision (1 if yes, 0 otherwise) [2016–2023]	1.410	12.25	–0.1949	0.1949
Roadway characteristics				
U-turn without auxiliary lane (1 if yes, 0 otherwise) [2016–2023]	0.369	2.74	–0.0506	0.0506
Environmental characteristics				
Raining (1 if yes, 0 otherwise) [2016–2023]	0.817	4.64	–0.1127	0.1127
Temporal characteristics				
Weekend (1 if yes, 0 otherwise) [2016–2023]	0.098	2.09	–0.0133	0.0133
Variables with temporally varying parameters				
Vehicle involvement				
One motorcycle involved (1 if yes, 0 otherwise) [2016–2020]	–0.599	–4.79	0.0818	–0.0818
One motorcycle involved (1 if yes, 0 otherwise) [2021–2023]	–0.276	–2.03	0.0384	–0.0384
One passenger car involved (1 if yes, 0 otherwise) [2016]	–0.475	–4.67	0.0688	–0.0688
One passenger car involved (1 if yes, 0 otherwise) [2017–2020]	–0.325	–4.49	0.0440	–0.0440
One passenger car involved (1 if yes, 0 otherwise) [2021–2023]	–0.653	–8.74	0.0921	–0.0921
One pickup truck involved (1 if yes, 0 otherwise) [2018–2021]	0.190	2.65	–0.0256	0.0256
Two or more pickup trucks involved (1 if yes, 0 otherwise) [2019–2023]	–0.535	–3.09	0.0724	–0.0724
One 6-wheel truck involved (1 if yes, 0 otherwise) [2016–2020]	1.282	8.88	–0.1747	0.1747
One 6-wheel truck involved (1 if yes, 0 otherwise) [2021–2023]	0.843	5.45	–0.1185	0.1185
Crash type				
Rear-end collision in the same lane (1 if yes, 0 otherwise) [2016–2018]	–0.295	–3.70	0.0391	–0.0391
Crash cause				
Speeding (1 if yes, 0 otherwise) [2016–2018]	–0.227	–2.77	0.0303	–0.0303
Speeding (1 if yes, 0 otherwise) [2023]	–0.238	–2.05	0.0341	–0.0341
Cutting in too closely (1 if yes, 0 otherwise) [2016–2022]	0.100	1.58	–0.0220	0.0220
(Standard deviation of parameter distribution; 52.9% above zero)	1.393	7.84		
Running a red light (1 if yes, 0 otherwise) [2016]	1.445	3.04	–0.2070	0.2070
Running a red light (1 if yes, 0 otherwise) [2018–2022]	0.880	6.14	–0.1202	0.1202
Roadway characteristics				
Crash on the main carriageway (1 if yes, 0 otherwise) [2016–2020]	–0.202	–1.70	0.0272	–0.0272
Crash on the main carriageway (1 if yes, 0 otherwise) [2021]	0.780	2.56	–0.1074	0.1074
Crash on a frontage road (1 if yes, 0 otherwise) [2016–2017]	–1.717	–7.25	0.2166	–0.2166
Crash on a frontage road (1 if yes, 0 otherwise) [2018–2020]	–0.803	–5.17	0.1059	–0.1059
Crash on a frontage road (1 if yes, 0 otherwise) [2022]	–1.790	–5.82	0.2211	–0.2211
Crash on a frontage road (1 if yes, 0 otherwise) [2023]	–0.787	–3.71	0.1108	–0.1108
Road under construction (1 if yes, 0 otherwise) [2023]	–1.008	–3.13	0.1369	–0.1369
Painted median (1 if yes, 0 otherwise) [2022–2023]	–0.351	–2.44	0.0482	–0.0482
Raised or earth median (1 if yes, 0 otherwise) [2016]	–0.860	–6.69	0.1206	–0.1206
Raised or earth median (1 if yes, 0 otherwise) [2017–2019]	–0.444	–5.14	0.0581	–0.0581

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Raised or earth median (1 if yes, 0 otherwise) [2021–2023]	–0.401	–4.86	0.0551	–0.0551
Depressed median (1 if yes, 0 otherwise) [2018–2023]	0.404	5.64	–0.0556	0.0556
Barrier median (1 if yes, 0 otherwise) [2016–2017]	–1.312	–8.31	0.1773	–0.1773
Barrier median (1 if yes, 0 otherwise) [2018–2023]	–1.020	–11.05	0.1368	–0.1368
Horizontal curve (1 if yes, 0 otherwise) [2016]	0.668	3.04	–0.0970	0.0970
Horizontal curve (1 if yes, 0 otherwise) [2017–2023]	0.392	3.29	–0.0540	0.0540
(Standard deviation of parameter distribution; 59.0% above zero)	1.719	4.92		
Concrete pavement surface (1 if yes, 0 otherwise) [2018]	–0.716	–4.32	0.0904	–0.0904
Concrete pavement surface (1 if yes, 0 otherwise) [2020–2021]	–0.718	–6.74	0.0993	–0.0993
T-intersection (1 if yes, 0 otherwise) [2016–2017]	–0.343	–2.07	0.0463	–0.0463
T-intersection (1 if yes, 0 otherwise) [2018]	0.450	1.85	–0.0566	0.0566
Four-leg intersection (1 if yes, 0 otherwise) [2018–2019]	0.544	2.68	–0.0710	0.0710
U-turn with deceleration lane only (1 if yes, 0 otherwise) [2016]	0.991	2.83	–0.1429	0.1429
U-turn with deceleration lane only (1 if yes, 0 otherwise) [2017–2023]	0.401	2.96	–0.0548	0.0548
U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise) [2016]	0.893	2.36	–0.1288	0.1288
U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise) [2021–2023]	0.868	3.28	–0.1214	0.1214
Environmental characteristics				
Wet road surface (1 if yes, 0 otherwise) [2019–2023]	–0.606	–2.69	0.0480	–0.0480
(Standard deviation of parameter distribution; 40.8% above zero)	2.611	4.49		
Dark, no street lighting (1 if yes, 0 otherwise) [2016–2021]	1.151	11.61	–0.1596	0.1596
Dark, no street lighting (1 if yes, 0 otherwise) [2022–2023]	0.533	2.82	–0.0753	0.0753
Dark, with street lighting (1 if yes, 0 otherwise) [2016–2018]	0.389	4.95	–0.0514	0.0514
Dark, with street lighting (1 if yes, 0 otherwise) [2020]	0.468	4.13	–0.0665	0.0665
Dark, with street lighting (1 if yes, 0 otherwise) [2021–2023]	0.244	2.35	–0.0338	0.0338
Temporal characteristics				
Songkran festival period (1 if yes, 0 otherwise) [2017–2023]	–0.767	–5.73	0.0544	–0.0544
(Standard deviation of parameter distribution; 40.6% above zero)	3.230	7.20		
New Year festival period (1 if yes, 0 otherwise) [2016–2018]	–0.683	–3.25	0.0271	–0.0271
(Standard deviation of parameter distribution; 44.2% above zero)	4.685	5.32		
New Year festival period (1 if yes, 0 otherwise) [2019–2023]	–1.056	–6.74	0.0864	–0.0864
(Standard deviation of parameter distribution; 36.2% above zero)	2.997	6.80		
Public holiday (1 if yes, 0 otherwise) [2022–2023]	0.439	1.72	–0.0617	0.0617
Evening (1 if yes, 0 otherwise) [2022–2023]	0.239	1.84	–0.0335	0.0335
Midnight / early morning (1 if yes, 0 otherwise) [2017–2023]	0.731	8.64	–0.1001	0.1001
Region				
Eastern region (1 if yes, 0 otherwise) [2016–2020]	1.376	12.04	–0.1885	0.1885
Eastern region (1 if yes, 0 otherwise) [2021–2023]	0.904	7.16	–0.1266	0.1266
Northeastern region (1 if yes, 0 otherwise) [2016–2017]	1.066	8.68	–0.1516	0.1516
Northeastern region (1 if yes, 0 otherwise) [2018–2023]	1.352	19.49	–0.1928	0.1928
Northern region (1 if yes, 0 otherwise) [2016–2019]	0.505	4.77	–0.0677	0.0677
Northern region (1 if yes, 0 otherwise) [2020–2022]	–0.366	–3.98	0.0505	–0.0505
Southern region (1 if yes, 0 otherwise) [2016–2017]	0.264	2.19	–0.0362	0.0362
Southern region (1 if yes, 0 otherwise) [2018–2023]	0.747	6.98	–0.1217	0.1217
(Standard deviation of parameter distribution; 67.1% above zero)	1.685	5.24		
Western region (1 if yes, 0 otherwise) [2016–2019]	0.899	6.94	–0.1210	0.1210
Western region (1 if yes, 0 otherwise) [2020]	1.283	5.99	–0.1877	0.1877
Western region (1 if yes, 0 otherwise) [2021–2023]	0.973	6.76	–0.1364	0.1364
Heterogeneity in the means of random parameters				
Southern region [2018–2023]: Rear-end collision in the same lane [2019–2023]	0.673	3.37		
Cutting in too closely [2016–2022]: Van involved [2016–2022]	1.154	3.22		
Cutting in too closely [2016–2022]: Crash on the main carriageway [2022–2023]	–2.086	–2.74		
Model statistics				

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Number of observations		21,371		
Number of estimated parameters		85		
Log-likelihood at zero, LL(0)		–14813.25		
Log-likelihood with constant only, LL(C)		–14745.16		
Log-likelihood at convergence, LL(β)		–11979.04		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$		0.191		
AIC / BIC		24,128.1 / 24,805.5		
Random parameter distribution; simulation		Normal; 2,000 Halton draws		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median. Periods in brackets share one parameter value; for a variable with temporally varying parameters, periods not shown have a parameter of zero. Temporal structure from the partially constrained procedure in the fixed-parameter binary logit (Stage 1): adjacent periods merged when a likelihood ratio test does not reject equal parameters at the 90% level. The stage at which the structure was chosen (fixed-parameter or random parameters model) is the one with the lower BIC among models not rejected at the 90% level against the fully unconstrained models (Table 3; all candidates in Table D1). Random parameters (at most eight), heterogeneity in means (at most three terms) and in variances (at most one term) were searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level (a parameter is kept random only while its standard deviation is significant). A variable used in a heterogeneity function enters only the random parameter shown, for crashes in the shared periods, and is removed from the main utility. Marginal effects are average changes in the simulated unconditional probability (2,000 Halton draws) when the indicator switches from 0 to 1, averaged over crashes in the parameter's periods; for random parameters they include the heterogeneity terms. PDO/minor = –(serious/fatal). Percentage above zero evaluated at the heterogeneity variables equal to zero.